

From sequence-centric modeling to topology-native pangenome pretraining

Input

Representation

Learning

Output / reuse

a Linear genomic foundation models

DNABERT, Nucleotide Transformer

linear reference or haplotype sequence

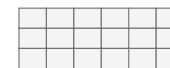


nucleotide tokens



pretrained sequence model (masked-nucleotide objective)

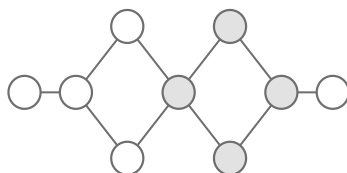
sequence embedding
→ downstream prediction



b Pangenome-derived sequence learning

DeepGene

population pangenome graph

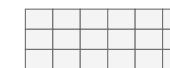


graph traversal & linearized to sequence



pretrained sequence model (transformer)

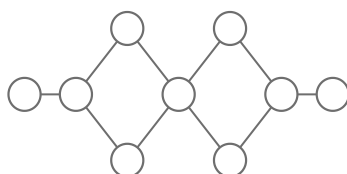
sequence embedding
→ downstream prediction



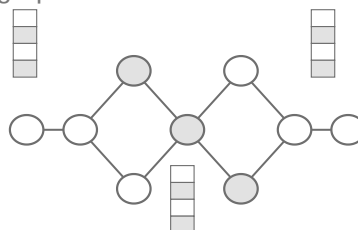
c Task-specific graph learning

PangenomeX

population pangenome graph



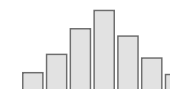
graph structure + node features



graph neural network (GNN)

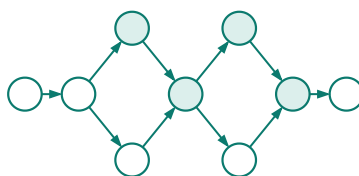
supervision (task labels)

CNV prediction



d PangenomeFM

native population-scale pangenome graph



oriented segments + genomic coordinates

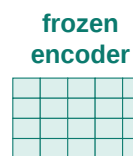
A +	1,254 - 1,689
B -	1,690 - 2,334
C +	2,335 - 2,912
D +	2,913 - 3,412

hide a queried relation, then reconstruct it



self-supervised topology pretraining reconstruct the hidden edge (no downstream labels)

frozen graph embedding



cross-graph transfer

cCRE classification

SV breakpoint classification